

Progressive Retrieval Attention on OpenVINO GenAI: Semantic Resource Identity, Continuous-Batching Evidence, and the Native K/V Boundary

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Abstract

OpenVINO GenAI provides continuous batching, bounded K/V scheduling, prefix caching, eviction controls, and execution across Intel CPU, GPU, and NPU devices. Progressive Retrieval Attention (PRA) adds semantic, query-addressed resources that must remain distinct from sequential prefixes. We implement an OpenVINO provider, an E0 selected-text adapter, and tested contracts for topology-bound native K/V handles, authorization, request pinning, exactly-once attachment, cleanup, and prefix-pool exclusion. A live audit of OpenVINO 2026.3.1 and OpenVINO GenAI 2026.3.1.0 confirms all required continuous-batching controls, but finds no public operation for attaching arbitrary selected non-prefix K/V. We therefore report native E2 as blocked, not as an implicit consequence of prefix caching. On Qwen2-0.5B-Instruct INT4, selected-text and full-context E0 recover the controlled answer in every evidence-bearing request on an Intel i7-1355U CPU and Iris Xe GPU. Selected text uses 63 input tokens instead of 1,573. On GPU it reduces cold TTFT from 1,129.6 to 102.9 ms. Under explicitly warmed continuous batching at eight shared requests, it reaches 8.48 requests/s rather than 7.39 and reduces TTFT p99 from 297.7 to 148.9 ms. An expanded ten-wave sweep reaches concurrency 16: selected context retains 100% controlled recovery and 188.9 output tokens/s, whereas shared full context follows a decoy in every request and reaches 141.1 output tokens/s. A separate 360-request natural cohort covers QASPER, HotpotQA, and 2WikiMultiHopQA; on Hotpot, selected context raises token F1 from 0.080 to 0.310 while cutting TTFT p99 from 2.50 s for full context to 387 ms. A matched Qwen2.5-1.5B INT4 replication adds another 360 requests. Selected/full F1 is 0.260/0.282 on QASPER and 0.332/0.346 on HotpotQA, while median full/selected TTFT ratios are 3.29 and 1.91. On 2Wiki the quality gap remains larger (0.157/0.259), localizing a context-coverage limitation rather than an engine transport failure. A cross-family TinyLlama-1.1B replication adds 360 more requests. Its selected/full F1 is 0.080/0.049 on QASPER, 0.090/0.110 on HotpotQA, and 0.083/0.080 on 2Wiki, while full context costs 1.71–4.18 times the selected median TTFT. The low absolute scores preclude a capability claim, but show that the transport advantage is not Qwen-tokenizer-specific. A second 149-example, 1,341-request natural-QA ablation holds the evidence fixed while adding one to eight answer-aware or low-overlap documents. QASPER F1 peaks at 0.265 with four answer-aware additions, versus 0.166 with evidence alone, but low-overlap additions reduce it to 0.140 at eight. A paired forensic audit shows that the QASPER gain is concentrated when added text contains answer terms; HotpotQA and 2Wiki do not show the same benefit. This is a diagnostic of lexical reinforcement and recency, not a deployable query-only selector. A counterbalanced 60-request physical-telemetry run on Iris Xe finds that selected E0 reduces median TTFT from 219.5 to 118.3 ms and OpenVINO-reported GPU allocation from 1,643 to 1,563 MiB. A 16-cell process-isolated lifecycle matrix compares OpenVINO GenAI 2026.2.1.0 with the latest 2026.3.1.0. Both releases pass short, one-shot long, long-to-short, and all cache-disabled cells, but the second cached long request violates cross-layer block release in both. Each teardown leaks one sequence block table and 59 of 128 blocks. The latest release therefore does not close this repeated-long prefix-cache/eviction defect. A three-point pressure curve keeps selection at 63 tokens while full context grows from 417 to 831 tokens; selected recovery remains exact throughout. The result establishes a reproducible transport baseline and a precise extension boundary; it does not establish native OpenVINO PRA parity or memory savings.

1 Qualification Snapshot

Engine question. Does PRA selection improve quality-adjusted Intel serving beyond OpenVINO’s prefix cache, and is detached native K/V available? The first answer is positive for the measured E0 workloads; the second remains blocked by the public attachment surface.

Table 1: Standard PRA engine result block. NOT MEASURED denotes a missing measurement, not zero cost or zero benefit.

Field	Result
Validated / observed level	E0 validated on CPU and GPU; E2 and E3 not observed
Model / hardware	Qwen2-0.5B, Qwen2.5-1.5B, and TinyLlama-1.1B INT4; i7-1355U and Iris Xe
Quality	Controlled success 1.00; paired natural selected F1 0.080–0.332; full F1 0.049–0.412
Visible context	63 selected tokens versus 417–1,573 full-context tokens
Loaded serving	11.1–14.2 selected requests/s and 233–330 ms TTFT p99 at concurrency 16
Native K/V tokens and bytes	NOT MEASURED: no public non-prefix attachment operation
Transfer / E2 reuse / E3 scheduling	NOT MEASURED
Evidence tier	Controlled CPU/GPU transport, three 360-request paired cohorts, a 1,341-request distractor cohort, 60 physical-telemetry requests, and a 16-cell lifecycle matrix

Recommendation. Use E0 selected text when frozen retrieval materially removes distractors, and evaluate successful tasks or output-token throughput beside raw requests/s. On this small natural cohort, selected evidence improves HotpotQA F1 and latency, but full context remains stronger on some datasets. Do not claim E2 until OpenVINO exposes selected non-prefix K/V to attention and a selector-frozen E0/E2 comparison establishes parity and reuse economics.

Primer. E0 renders the selected evidence as ordinary prompt text. E1 preserves PRA resource identities, typed records, routing, and fallback policy through the runtime while attention remains ordinary. E2 exposes detached selected native K/V directly to attention. E3 additionally makes their physical placement, sharing, promotion, and eviction scheduler-owned. This paper validates E0 and specifies, but does not measure, the native levels.

2 Introduction

An inference engine’s prefix cache identifies previously computed sequential token histories. PRA identifies immutable logical resources selected by a query. These identities have different lifetimes and authorization rules:

$$I_{\text{prefix}} \neq I_{\text{PRA}} \neq I_{\text{physical}}.$$

Identical visible prompts may select different protected resources, while the same resource may be shared safely by many sessions. A correct OpenVINO integration must preserve that distinction while allowing OpenVINO to keep control of compiled execution, K/V blocks, eviction, and batching.

This study makes three contributions. It implements the PRA-side OpenVINO contracts, audits the installed GenAI scheduler and pipeline interfaces, and measures a common E0 workload across CPU, GPU, prefix reuse, and continuous batching. The audit also identifies the smallest missing native mechanism: request attention metadata cannot currently receive an arbitrary selected non-prefix K/V set through the public Python API.

3 Claim Boundary

E0 renders selected source text into the visible prompt. E1 carries PRA logical identities, typed records, routing decisions, and explicit fallback policy while model attention remains ordinary. E2 consumes selected detached K/V natively, preserving source geometry and combining local and memory scores in one softmax. E3 gives the engine scheduler physical ownership of native PRA placement, sharing, promotion, eviction, and accounting. All model measurements in this paper are E0. The native classes described below are executable lifecycle contracts, not evidence that OpenVINO already performs E2 or E3.

4 OpenVINO Mapping

4.1 Responsibility split

PRA owns typed records, selection, authorization, task/session policy, semantic retention, and a four-tier storage lifecycle. OpenVINO owns compilation, device execution, sequential K/V blocks, prefix reuse, batching, and physical eviction. Physical tensor handles are derived state bound to a specific model, revision, device, precision, layer count, K/V-head count, and head dimension.

4.2 Implemented code

`src/pras/openvino/adapters.py` provides `OpenVINOEngineAdapter`. Its HTTP/OVMS form is explicitly E0. It reports E2 only when an application supplies an `OpenVINONativeExecutor`; package installation or prefix-cache availability is never enough.

`src/pras/openvino/native.py` provides a topology fingerprint, immutable K/V handles, a physical-store protocol, an in-memory deterministic store, and `OpenVINONativeAttachmentManager`. The manager validates topology before registration, authorizes a complete routed set before scheduler admission, pins it for the request lifetime, returns it exactly once, and unpins it during cleanup. A shared isolation guard rejects any PRA logical key offered to the ordinary sequential/prefix pool. Focused tests cover topology mismatch, cross-tenant access, duplicate attachment, cleanup, and capability honesty.

4.3 Live API boundary

`SchedulerConfig` exposes cache size or K/V-block count, sequence and batched-token limits, prefix caching, dynamic split/fuse, and cache eviction. Supplying this configuration activates continuous batching. The public `LLMPipeline` surface exposes generation, tokenizer/config access, and chat state. It does not expose selected external K/V, non-prefix tensor attachment, or an attention-metadata callback. Prefix caching therefore accelerates repeated visible histories but does not implement PRA E2.

5 Experimental Method

5.1 Environment

The remote Windows host runs Python 3.10.0, OpenVINO 2026.3.1, OpenVINO GenAI 2026.3.1.0, and OpenVINO Tokenizers 2026.3.1.0. Available devices are a 13th Gen Intel Core i7-1355U CPU and Intel Iris Xe integrated GPU; no NPU is available. The first model is OpenVINO Qwen2-0.5B-Instruct INT4. Cross-model replication uses Qwen2.5-1.5B-Instruct and TinyLlama-1.1B-Chat-v1.0, exported with Optimum-Intel 2.1.0 and OpenVINO 2026.3.1 using `optimum-cli export openvino --weight-format int4 --group-size 128`. Scheduler settings use a 1 GiB cache, eight sequences, 2,048 batched tokens, prefix caching, and dynamic split/fuse. Both paired natural cohorts use 20 examples per dataset, two repeats, a 384-token selected cap, a 1,536-token full cap, and at most 24 greedy output tokens.

5.2 Conditions

The target resource contains PRA_EVIDENCE_4821; distractors contain different codes. The five one-shot conditions are no evidence, stable prefix only, selected text, prefix plus selected text, and full retained context. The same evidence is used in selected and full conditions. Each condition is run five times independently on CPU and GPU.

The direct `ContinuousBatchingPipeline` experiment uses selected text and full context under shared-resource and independent-resource workloads. Concurrency is 1, 2, 4, and 8 with five measured waves. Every distinct resource is primed outside the timed interval. Reported p95/p99 values are smoke descriptors over small cohorts, not production service-level estimates. The expanded campaign uses ten waves at concurrency 1, 2, 4, 8, and 16, a 4,096-token batch budget, and six repeated distractors. It records success, requests/s, output tokens/s, TTFT, TPOT, generation time, input tokens, and RSS.

The distractor ablation uses 149 examples: 49 QASPER, 50 HotpotQA, and 50 2WikiMultiHopQA. The evidence-only prompt and eight additions are constructed once per example. Both addition orders draw from the same example’s nonselected candidate blocks. The condition previously called “relevant” ranks blocks by lexical overlap with the question *and gold answer*; the “irrelevant” condition takes the lowest-overlap blocks. Added blocks follow the selected evidence and precede the question. Conditions add 1, 2, 4, or 8 blocks while preserving the answer-bearing evidence; later additions, rather than selected evidence, are truncated at the input cap. One greedy GPU request per condition yields 1,341 requests. We report token F1, exact match, answer containment, source-token count, and TTFT. Gold answer log-probability is NOT MEASURED by the public pipeline.

6 Results

6.1 Capability status

Table 2: OpenVINO integration status after live package and hardware audit.

Capability	Status	Evidence
Selected-text E0	Ready	CPU/GPU live generation
Prefix cache controls	Ready	live SchedulerConfig audit
Continuous batching	Ready	direct batched generation
PRA lifecycle/isolation	Contract ready	focused tests
Native selected K/V E2	Blocked	no public attachment operation
OVMS E0	Adapter ready	endpoint not installed on host
NPU comparison	Not run	no NPU device available

6.2 CPU and GPU one-shot behavior

Table 3: Evidence-bearing E0 conditions. Visible is mean input tokens; times are milliseconds. Success is controlled answer recovery.

Device	condition	visible	success	cold TTFT	warm TTFT	latency p50
CPU	pra_only	63	1.00	549.2	336.8	1641.2
CPU	prefix_plus_pra	361	1.00	909.8	191.2	1118.1
CPU	full_context	1573	1.00	7600.3	180.6	1249.1
GPU	pra_only	63	1.00	102.9	99.1	788.3
GPU	prefix_plus_pra	361	1.00	115.0	98.5	814.6
GPU	full_context	1573	1.00	1129.6	98.2	501.3

Controls recover the answer in 0/10 requests across CPU and GPU; every evidence-bearing condition succeeds in 30/30. Selected text uses 63 tokens, 96.0% fewer than the 1,573-token full context. On CPU, cold TTFT falls from 7,600.3 ms for full context to 549.2 ms for selected text. On GPU it falls from 1,129.6 to 102.9 ms. Prefix caching changes the comparison: warm full-context TTFT is 180.6 ms on CPU and 98.2 ms on GPU, versus 336.8 and 99.1 ms for selected text. Thus selected text strongly reduces cold ingestion, while a fully reusable prefix can remove most repeated-prefill cost in this fixed one-resource smoke.

Pipeline construction is reported separately: 1.58 s for the CPU one-shot process and 39.40 s for the first GPU one-shot process. The later direct continuous-batching GPU process constructs in 5.05 s after system/model caches have been populated; these are cold-process observations, not comparable steady-state request latencies.

6.3 Cache-admission and lifecycle boundary

We then enable cache eviction on Qwen2.5-1.5B and constrain the scheduler by physical K/V blocks. The benchmark now distinguishes a capacity rejection from an internal cache-lifecycle failure, preserves partial rows, and stops issuing requests after a terminal reference-count violation.

Table 4: OpenVINO GenAI cache boundary. Selected is visible selected text, not native E2. Full n reports measured requests out of ten; tail metrics exclude all rejected or post-corruption cells.

K/V blocks	Selected n	Selected success	Selected warm TTFT	Full n	Full status	Full cold TTFT
64	10	1.00	152.5	0/10	cache admission	–
128	10	1.00	141.4	1/10	lifecycle abort	4,715.3

At 64 blocks, all 40 reduced-context requests complete before full context is rejected ten times for insufficient cache budget. This is a normal admission boundary. At 128 blocks, all four reduced conditions again complete 10/10 and the first 1,573-token full-context request succeeds. The next full request fails the engine invariant `m_ref_count > 0`; teardown reports a leaked sequence block table and 58 unreleased blocks. A 256-block diagnostic reproduces the invariant failure, so additional capacity does not cure it.

The result does not weaken selected-text correctness. It identifies a prefix-cache/eviction lifecycle defect in OpenVINO GenAI 2026.3.1.0 under this repeated long-prefix workload. The harness labels subsequent cells `NOT_RUN.CACHE.LIFECYCLE`; it never converts them to zero latency or continues timing a corrupted pipeline. Production use should disable this cache combination or upgrade to a release that passes the same regression.

To distinguish a generic long-context failure from a state transition, we ran a second matrix in fresh processes. It crosses OpenVINO GenAI 2026.2.1.0 and the latest PyPI release, 2026.3.1.0, with cache enabled or disabled and four ordered workloads: repeated short, one long request, long followed by five short requests, and five repeated long requests. The cache-enabled condition sets both prefix caching and cache eviction; the disabled control turns both off. All cells use the same Qwen2.5-1.5B INT4 model, Iris Xe device, 128-block budget, and deterministic generation.

Table 5: Process-isolated lifecycle regression. Completed is measured requests over planned requests. Quality means every completed request recovered the controlled answer. Leaked is the teardown block deficit.

GenAI	Cache	Workload	Status	Completed	Quality	Leaked
2026.2.1.0	disabled	long once	measured	1/1	yes	–
2026.2.1.0	disabled	long → long	measured	5/5	yes	–
2026.2.1.0	disabled	long → short	measured	6/6	yes	–
2026.2.1.0	disabled	short repeated	measured	5/5	yes	–
2026.2.1.0	enabled	long once	measured	1/1	yes	–
2026.2.1.0	enabled	long → long	lifecycle	1/5	yes	59
2026.2.1.0	enabled	long → short	measured	6/6	yes	–
2026.2.1.0	enabled	short repeated	measured	5/5	yes	–
2026.3.1.0	disabled	long once	measured	1/1	yes	–
2026.3.1.0	disabled	long → long	measured	5/5	yes	–
2026.3.1.0	disabled	long → short	measured	6/6	yes	–
2026.3.1.0	disabled	short repeated	measured	5/5	yes	–
2026.3.1.0	enabled	long once	measured	1/1	yes	–
2026.3.1.0	enabled	long → long	lifecycle	1/5	yes	59
2026.3.1.0	enabled	long → short	measured	6/6	yes	–
2026.3.1.0	enabled	short repeated	measured	5/5	yes	–

The result localizes the defect. Both versions complete 5/5 repeated short, 1/1 one-shot long, and 6/6 long-to-short requests with the cache enabled. Both also complete every cache-disabled sequence, including 5/5 repeated long. In contrast, the first cached long request succeeds and the second fails `is_all_free`: blocks across layers must be freed simultaneously. Teardown then reports one leaked sequence block table and 69 free blocks where 128 are expected, a 59-block deficit repeated by 28 allocator reports. All 60 requests that reached generation recovered the expected answer.

This controlled comparison changes the operational advice: upgrading from 2026.2.1.0 to 2026.3.1.0 is not a remedy. For this model and scheduler geometry, repeated long-prefix production should avoid the combined prefix-cache and eviction configuration until a later release passes this exact regression. The failure belongs to the engine cache lifecycle, not to PRA routing or selected-text semantics.

6.4 Continuous batching

Table 6: Explicitly primed GPU continuous-batching endpoints. Every row has controlled answer success 1.00. RSS is process resident memory after the group.

Transport / resources	C	req/s	TTFT p99	generation p99	RSS MiB
selected / shared	1	1.35	159.7	791.6	2140
selected / shared	8	8.48	148.9	957.8	1950
selected / independent	1	1.57	115.6	760.3	1960
selected / independent	8	5.88	262.0	1430.8	2147
full / shared	1	1.55	223.7	832.1	2163
full / shared	8	7.39	297.7	1154.7	2115
full / independent	1	1.46	188.3	850.3	2119
full / independent	8	7.07	319.8	1192.3	2193

With one shared resource and concurrency eight, selected text reaches 8.48 requests/s compared with 7.39 for full context, while TTFT p99 falls from 297.7 to 148.9 ms. For independent resources, selected

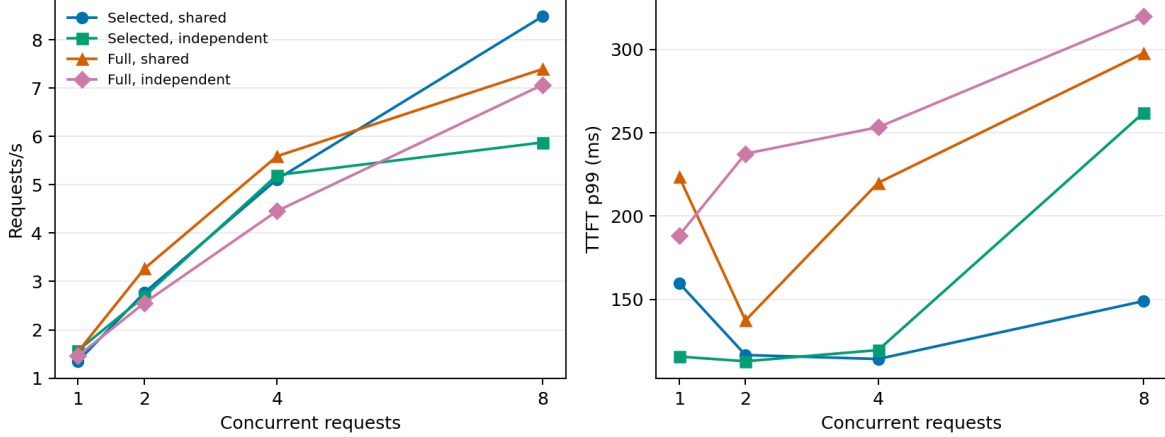


Figure 1: Warm OpenVINO GenAI E0 throughput and TTFT tail under direct continuous batching. Selected denotes rendered selected text, not native E2 K/V.

text still lowers TTFT p99 (262.0 versus 319.8 ms) but reaches only 5.88 requests/s versus 7.07. This small reversal prevents a blanket throughput claim. It may reflect output length, scheduler packing, and churn in the explicitly primed independent set; larger cohorts and matched generated-token work are required.

RSS spans roughly 1.95–2.19 GiB across endpoint rows and is not monotonic with transport or concurrency. The current experiment therefore supports a token and latency claim, not an engine-memory saving claim.

6.5 Natural QA and expanded continuous batching

Table 7: Natural-QA E0 cohort: 20 examples per dataset and two repeats. Times are milliseconds; retrieval is frozen across the execution conditions.

Dataset	Context	F1	Contain.	Prompt tok.	TTFT p50	TTFT p95
2Wiki	None	0.078	0.150	39	180.5	237.9
2Wiki	Selected	0.121	0.275	387	234.8	389.8
2Wiki	Full	0.137	0.275	877	267.7	1887.8
hotpotqa	None	0.080	0.050	45	179.1	253.3
hotpotqa	Selected	0.310	0.350	428	277.8	378.6
hotpotqa	Full	0.412	0.475	1339	527.0	2432.4
qasper	None	0.066	0.050	34	168.4	284.1
qasper	Selected	0.124	0.050	411	241.4	384.6
qasper	Full	0.219	0.250	1561	705.5	2737.9

The Qwen2-0.5B INT4 checkpoint shows useful but incomplete QA competence. Selected context improves token F1 over no context on all datasets: 0.121 versus 0.078 on 2Wiki, 0.310 versus 0.080 on HotpotQA, and 0.124 versus 0.066 on QASPER. Full context is stronger in this bounded cohort, with F1 0.137, 0.412, and 0.219, exposing a retrieval/evidence-coverage gap rather than a transport failure. Selected median TTFT is 235–278 ms, compared with 268–705 ms for full context. The tail difference is larger: selected p95 is 379–390 ms, whereas full-context p95 is 1.89–2.74 s.

6.6 Cross-model natural replication

Table 8: Matched natural E0 results across two Qwen sizes and a Llama-family checkpoint. Each cell aggregates 20 unique examples and two repeats. TTFT is median milliseconds; the ratio is full divided by selected, so larger favors selected transport.

Model	Dataset	Sel. F1	Full F1	Sel. TTFT	Full TTFT	TTFT ratio
Qwen2 0.5B	2Wiki	0.121	0.137	235	268	1.14×
Qwen2 0.5B	HotpotQA	0.310	0.412	278	527	1.90×
Qwen2 0.5B	QASPER	0.124	0.219	241	705	2.92×
Qwen2.5 1.5B	2Wiki	0.157	0.259	483	635	1.31×
Qwen2.5 1.5B	HotpotQA	0.332	0.346	609	1164	1.91×
Qwen2.5 1.5B	QASPER	0.260	0.282	486	1597	3.29×
TinyLlama 1.1B	2Wiki	0.083	0.080	269	459	1.71×
TinyLlama 1.1B	HotpotQA	0.090	0.110	391	1211	3.10×
TinyLlama 1.1B	QASPER	0.080	0.049	327	1367	4.18×

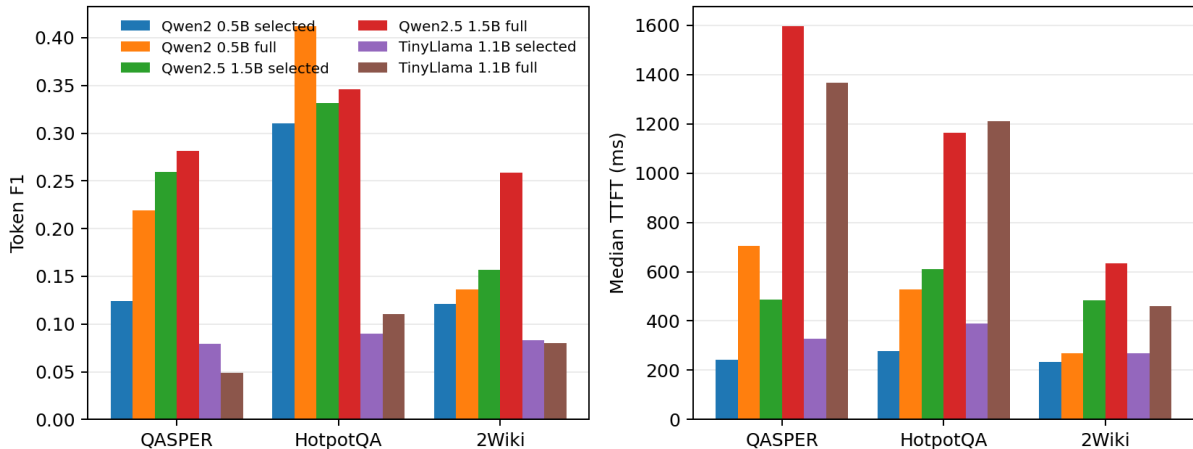


Figure 2: Cross-model quality and TTFT with frozen evidence selection. Model capacity and family change the quality response, while selected transport is faster in all nine matched dataset/model comparisons.

The 1.5B checkpoint raises selected F1 from 0.124 to 0.260 on QASPER, from 0.310 to 0.332 on HotpotQA, and from 0.121 to 0.157 on 2Wiki. More importantly, the selected/full gap narrows from 0.095 to 0.022 on QASPER and from 0.102 to 0.014 on HotpotQA. This is positive cross-model evidence for bounded selected context: a more capable model uses the frozen selection nearly as well as the larger visible prompt on two datasets.

The exception is informative. On 2Wiki the 1.5B selected/full gap is 0.102, versus 0.016 at 0.5B, while full-context F1 rises to 0.259. The result is consistent with missing multi-hop coverage in the fixed selection; it is not evidence against E0 transport. Across the 1.5B rows, full/selected median TTFT is 3.29 on QASPER, 1.91 on HotpotQA, and 1.31 on 2Wiki. Full-context p95 TTFT reaches 3.71–6.30 s, versus 1.21–1.29 s selected. Thus model scaling preserves the input-latency advantage while making retrieval coverage, rather than decoder capacity alone, the remaining quality variable.

TinyLlama provides a deliberately different family and tokenizer. Its low absolute F1 (0.049–0.110 across selected and full conditions) does not support an answer-quality claim. It does provide a useful transport replication: selected median TTFT is 269–391 ms, full is 459–1,367 ms, and selected is faster in every dataset. Selected slightly exceeds full F1 on QASPER and 2Wiki, but trails by 0.021 on HotpotQA. Together, the three models show that additional visible context is not uniformly beneficial and that the latency cost of presenting it survives a change in model family.

6.7 Natural distractor sensitivity

Table 9: Natural-QA token F1 after adding answer-aware high-overlap (Rel.) or low-overlap (Irr.) blocks to fixed answer-bearing evidence. Each cell aggregates 49 QASPER or 50 HotpotQA/2Wiki examples.

Dataset	Evidence	Rel. 1	Rel. 2	Rel. 4	Rel. 8	Irr. 1	Irr. 2	Irr. 4	Irr. 8
QASPER	0.166	0.217	0.231	0.265	0.205	0.191	0.162	0.170	0.140
HotpotQA	0.297	0.248	0.265	0.289	0.251	0.234	0.238	0.218	0.201
2Wiki	0.155	0.132	0.133	0.109	0.107	0.119	0.133	0.134	0.158

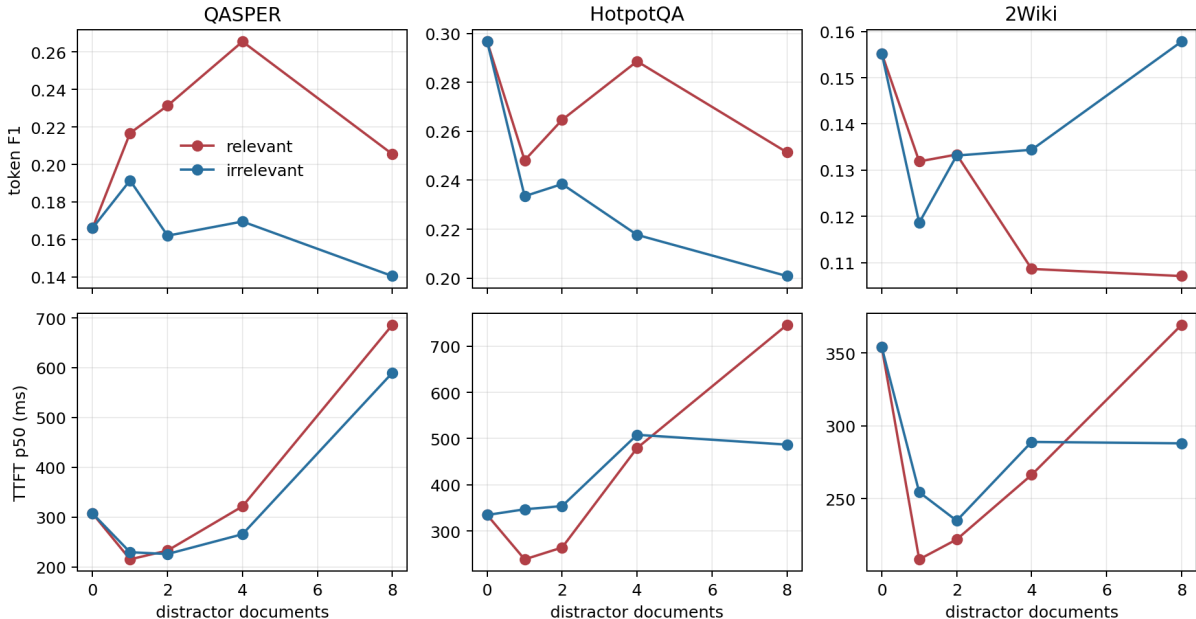


Figure 3: Quality and TTFT under context dilution. The point at zero is the same evidence-only condition for both curves. The selector is frozen; only the added documents differ.

The response is dataset dependent. On QASPER, four answer-aware additions raise F1 from 0.166 to 0.265, a 59.8% relative gain, before falling to 0.205 at eight. Eight irrelevant additions instead reduce F1 to 0.140. This is a bounded coverage benefit: some related context helps, but adding everything does not. HotpotQA starts strongest with evidence alone (0.297). Four relevant additions nearly recover that level (0.289), while eight irrelevant additions fall to 0.201. On 2Wiki, relevant additions decline from 0.155 to 0.107; the irrelevant curve is non-monotonic and ends at 0.158. Given the small model and 50-example cells, the 2Wiki reversal is noise-compatible and is not evidence that irrelevant context helps.

Latency generally increases at the largest context. At eight relevant documents, median TTFT is 686 ms on QASPER, 745 ms on HotpotQA, and 369 ms on 2Wiki, compared with evidence-only medians of 307, 337, and 354 ms. Smaller conditions fluctuate because compilation is already warm, sequence lengths and generated lengths differ, and the integrated GPU shares host resources. The robust conclusion is therefore about the joint frontier: evidence coverage and dilution must be calibrated per workload, while unbounded visible context consumes more input work without a reliable quality gain.

The post-hoc paired audit in Table 10 explains the apparent QASPER exception. At four answer-aware additions, mean paired F1 rises by 0.099: 40.8% of examples improve and 20.4% are harmed. The exact normalized answer appears in an added block for 30.6% of examples. When it appears, mean F1 rises by 0.147; without it, the gain is only 0.030. By contrast, answer-absent additions reduce F1 by 0.073 on

HotpotQA and 0.057 on 2Wiki. Across all paired rows, F1 change correlates weakly positively with answer-term overlap ($r = 0.171$), negatively with question overlap ($r = -0.104$), and negligibly with added-token count ($r = -0.024$).

Table 10: Paired distractor forensics. “Answer present” is the fraction whose added text contains the exact normalized answer; “cap hit” records truncation of later additions. Rel. uses question-plus-gold-answer overlap and is not a deployable selector.

Dataset	order	k	$\Delta F1$	improved	harmd	answer present	cap hit
2wikimultihopqa	irrelevant	1	-0.037	0.08	0.14	0.12	0.00
2wikimultihopqa	irrelevant	2	-0.022	0.08	0.12	0.22	0.00
2wikimultihopqa	irrelevant	4	-0.021	0.16	0.16	0.32	0.00
2wikimultihopqa	irrelevant	8	+0.003	0.12	0.12	0.50	0.06
2wikimultihopqa	relevant	1	-0.023	0.08	0.12	0.12	0.00
2wikimultihopqa	relevant	2	-0.022	0.06	0.12	0.20	0.00
2wikimultihopqa	relevant	4	-0.047	0.10	0.20	0.34	0.02
2wikimultihopqa	relevant	8	-0.048	0.08	0.16	0.50	0.06
hotpotqa	irrelevant	1	-0.063	0.16	0.18	0.04	0.00
hotpotqa	irrelevant	2	-0.058	0.08	0.18	0.08	0.00
hotpotqa	irrelevant	4	-0.079	0.12	0.26	0.20	0.06
hotpotqa	irrelevant	8	-0.096	0.22	0.32	0.44	0.14
hotpotqa	relevant	1	-0.048	0.16	0.18	0.24	0.00
hotpotqa	relevant	2	-0.032	0.18	0.18	0.32	0.00
hotpotqa	relevant	4	-0.008	0.24	0.18	0.42	0.00
hotpotqa	relevant	8	-0.045	0.18	0.26	0.44	0.14
qasper	irrelevant	1	+0.025	0.29	0.16	0.02	0.00
qasper	irrelevant	2	-0.004	0.20	0.14	0.02	0.00
qasper	irrelevant	4	+0.003	0.29	0.14	0.02	0.00
qasper	irrelevant	8	-0.026	0.29	0.24	0.02	0.10
qasper	relevant	1	+0.051	0.41	0.16	0.24	0.00
qasper	relevant	2	+0.065	0.39	0.18	0.29	0.00
qasper	relevant	4	+0.099	0.41	0.20	0.31	0.02
qasper	relevant	8	+0.039	0.35	0.18	0.33	0.24

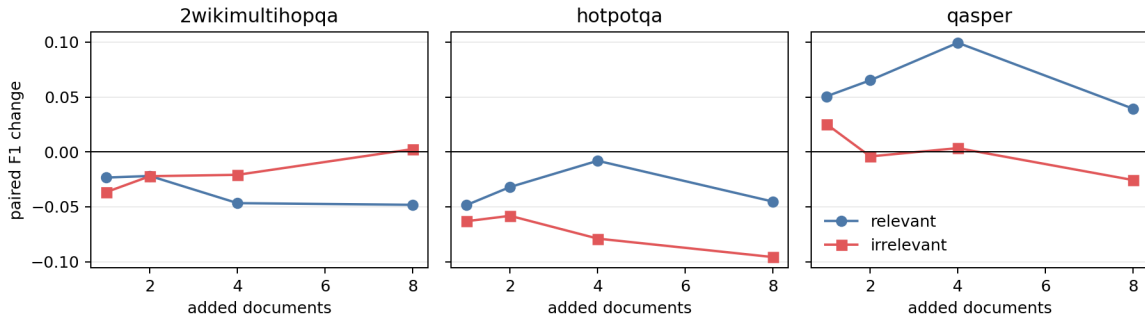


Figure 4: Per-example quality changes expose lexical reinforcement rather than a generic benefit from adding related context. QASPER’s positive tail is strongest when answer strings recur near the question; the two multi-hop datasets do not share that response.

6.8 Physical Intel telemetry

The counterbalanced physical run alternates selected and full conditions by example and repeat, for 30 requests per condition on the Iris Xe GPU. Selected E0 carries 348 source tokens on average; FULL E0

carries 1,227. Table 11 reports process counters and the plugin’s GPU_MEMORY_STATISTICS allocation counter. It does not reinterpret process RSS as device residency.

Table 11: Counterbalanced Intel physical telemetry. GPU MiB is OpenVINO plugin allocation; RSS and process I/O come from the operating system. Energy and GPU utilization are not measured because the host exposes no nonprivileged counter.

Condition	source tokens	F1	TTFT (ms)		GPU MiB peak	RSS MiB peak	CPU s mean	read KiB mean
			p50	p95				
selected E0	348.0	0.159	118.3	246.6	1563.1	2486.8	1.24	219.9
FULL E0	1226.9	0.127	219.5	1298.9	1643.1	2471.6	1.94	428.5

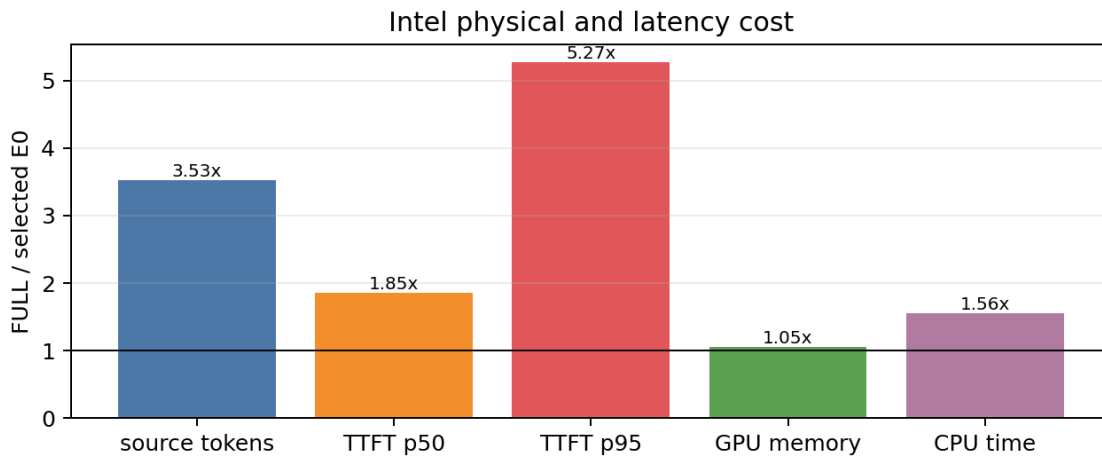


Figure 5: FULL/selected ratios from the live Intel run. Full context increases median TTFT by 1.85 $\times$, p95 TTFT by 5.27 $\times$, process CPU by 1.56 $\times$, bytes read by 1.95 $\times$, and plugin allocation by 1.05 $\times$.

Selected E0 also scores 0.159 F1 versus 0.127 for FULL E0 in this 60-request slice. The physical result therefore reinforces the E0 quality-adjusted frontier: 3.53 $\times$ more source tokens do not buy better answers, while they raise median completion latency by 19.4% and p95 latency by 86.0%. It remains an E0 comparison; no physical native-K/V saving is inferred.

Table 12: Expanded GPU batching endpoints. Each row has ten waves. Success and output-token throughput are shown because wrong short answers can inflate raw requests/s.

Workload	Context	C	Succ.	Req/s	Out tok/s	TTFT p50	TTFT p99	ITL p99
Shared	Selected	1	1.00	1.45	24.7	101.7	126.1	38.2
Shared	Selected	8	1.00	8.33	141.6	137.5	157.7	6.6
Shared	Selected	16	1.00	11.11	188.9	297.7	329.8	5.3
Independent	Selected	1	1.00	1.33	19.4	120.4	148.8	60.8
Independent	Selected	8	1.00	5.38	82.6	231.7	280.6	12.7
Independent	Selected	16	1.00	8.36	124.9	354.7	440.3	7.8
Shared	Full	1	0.00	1.68	16.8	113.9	203.7	69.1
Shared	Full	8	0.00	8.23	82.3	216.0	302.4	11.8
Shared	Full	16	0.00	14.11	141.1	317.6	351.2	5.8
Independent	Full	1	0.30	1.64	17.0	126.1	192.9	70.1
Independent	Full	8	0.51	7.05	75.1	220.8	301.3	13.8
Independent	Full	16	0.45	12.21	129.3	282.3	375.1	6.7

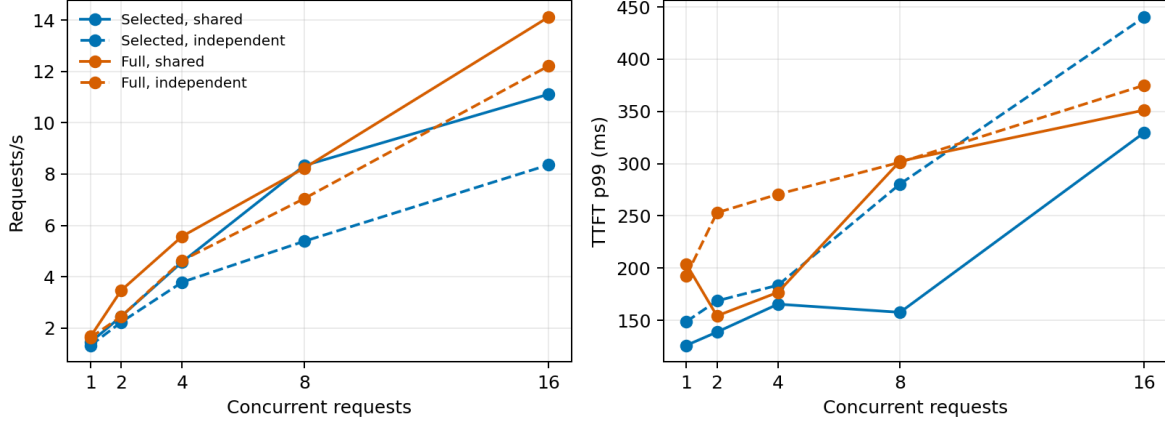


Figure 6: Expanded OpenVINO E0 throughput and TTFT tails through concurrency 16. The full-context request/s lines include wrong, shorter completions and must be read with Table 12.

Selected context recovers the target in every expanded request. At shared concurrency 16 it reaches 11.1 requests/s and 188.9 output tokens/s, compared with 14.1 requests/s and 141.1 output tokens/s for full context. The latter follows a distractor in all 160 requests and terminates after ten tokens rather than 17. Under independent resources, selected remains exact while full context success ranges from 0.30 to 0.51. At concurrency 16, selected reaches 8.36 requests/s, 124.9 output tokens/s, and 440 ms TTFT p99; full reaches 12.2, 129.3, and 375 ms but only 0.45 success. This campaign therefore establishes a quality-adjusted selected-context advantage, not a blanket latency win.

6.9 Context-pressure curve

To separate concurrency from retained-context size, we repeat the peak-load endpoint at three frozen prompt constructions. The selected condition is unchanged at 63 input tokens; the full condition grows from 417 to 677 and 831 tokens. Each point uses concurrency 16 and ten measured waves after explicit warmup.

Table 13: OpenVINO context pressure at concurrency 16. Success is reported beside throughput because the model sometimes follows a distractor and emits a shorter answer.

Size	Workload	Context	Prompt tok.	Succ.	Req/s	Out tok/s	TTFT p99
Small	Shared	Selected	63	1.00	14.16	240.7	233.1
Small	Independent	Selected	63	1.00	12.75	188.1	242.7
Small	Shared	Full	417	1.00	11.14	267.4	168.5
Small	Independent	Full	417	0.88	8.15	153.0	246.5
Medium	Shared	Selected	63	1.00	11.11	188.9	329.8
Medium	Independent	Selected	63	1.00	8.36	124.9	440.3
Medium	Shared	Full	677	0.00	14.11	141.1	351.2
Medium	Independent	Full	677	0.45	12.21	129.3	375.1
Large	Shared	Selected	63	1.00	14.12	240.0	234.8
Large	Independent	Selected	63	1.00	10.81	161.1	318.6
Large	Shared	Full	831	1.00	11.52	126.7	483.3
Large	Independent	Full	831	0.82	10.18	112.6	529.7

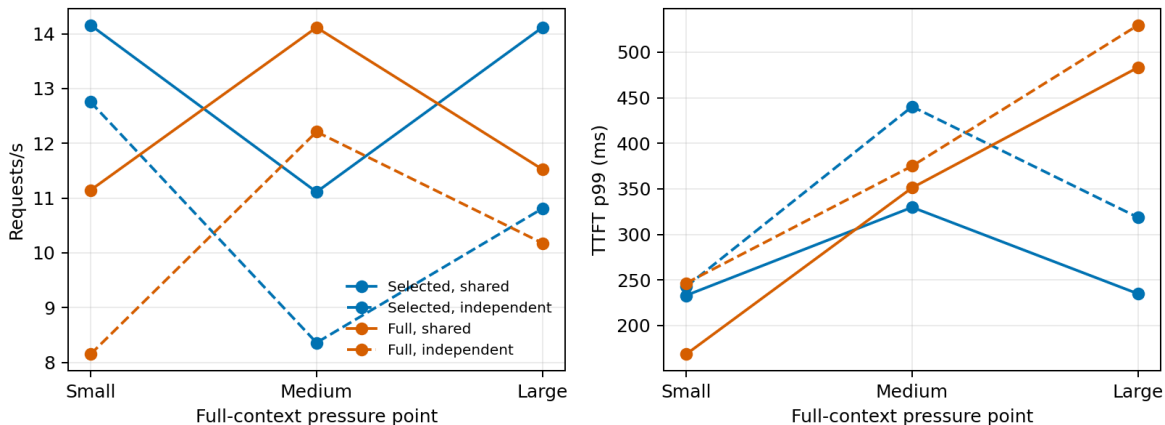


Figure 7: Peak-load throughput and TTFT tail as full visible context grows. Selected context remains fixed; solid lines share one resource and dashed lines use independent resource identities.

Selected evidence remains exact at all three pressure points. Its shared throughput ranges from 11.1 to 14.2 requests/s and TTFT p99 from 233 to 330 ms without increasing visible input. Full-context TTFT p99 grows from 169 ms at 417 tokens to 483 ms at 831 tokens for a shared resource, and from 247 to 530 ms for independent resources. Full-context correctness is not monotonic: it is exact at the smallest and largest shared points but follows a decoy throughout the middle shared point. This is a prompt-order/model effect, not evidence loss, and prevents interpreting raw throughput as a monotonic quality curve. The stable result is narrower: selection bounds input work and preserves the controlled target as retained context grows.

7 Interpretation

OpenVINO prefix caching and PRA selection address different reuse axes. When the same full context is repeatedly visible, prefix caching can make its warm TTFT competitive with selected text. When requests arrive concurrently, shorter selected inputs reduce scheduler work and improve the shared-resource frontier in this smoke. Native PRA would target a harder regime: encode a resource once, select different non-prefix intervals for many queries, and avoid making those source tokens visible. That regime cannot be measured until the attention attachment seam exists.

8 Required Native Extension

A minimal extension should accept an authorized set of topology-matched K/V handles plus source positions at request admission. The scheduler must pin the set, keep it outside ordinary prefix identity, expose it once to attention, and release it on completion or cancellation. Attention must normalize local and memory scores jointly. A geometry ladder should then compare ordinary prefix, geometry-matched cached source, full native source, sparse multi-record source, and cached decode before any economics claim.

9 Limitations

The controlled task verifies evidence transport; each paired natural cohort has only 20 unique examples per dataset and the distractor ablation has 49–50 per dataset with one generation per condition. Three small INT4 models from two families, one CPU, one integrated GPU, and at most ten waves per load endpoint are measured. TinyLlama’s weak task scores make it a transport check rather than a strong quality replication. There is no OVMS process, NPU, native K/V parity, energy or GPU-utilization telemetry, gold answer log-probability, long-session pressure, eviction sweep, or larger capable cross-family replication. Plugin allocation is available, but physical residency by cache class is not. The lifecycle version matrix covers two adjacent

release lines, one model, one GPU, one 128-block geometry, and five repeats; it localizes a deterministic failure but does not estimate its prevalence across models or devices. All serving results are E0.

10 Conclusion

PRA can coexist with OpenVINO’s public serving abstractions at E0 while preserving semantic identity and explicit capability reporting. Selected text substantially reduces cold ingestion and natural-input TTFT, and it avoids the decoy failures seen in the expanded full-context control. OpenVINO GenAI exposes the physical scheduling controls needed around a future integration but not the public arbitrary non-prefix K/V attachment needed for E2. That boundary is now implemented as a safe contract and measured as an open gate. The 1.5B replication shows that selected context approaches full-context quality on QASPER and HotpotQA while retaining lower TTFT, but 2Wiki keeps a meaningful coverage gap; better routing remains part of the product requirement. TinyLlama confirms the selected-input latency advantage across a second family without being capable enough to settle answer quality. The distractor audit narrows the QASPER gain to answer-aware lexical reinforcement rather than generic context breadth. Counterbalanced Intel telemetry further shows that the selected E0 frontier reduces TTFT, completion tails, process work, and plugin allocation in the measured slice. The isolated lifecycle matrix further shows that the cached repeated-long defect persists in the latest tested release, while one-shot long, long-to-short, and cache-disabled controls remain healthy. OpenVINO remains primarily an E0 result until a legitimate detached-K/V seam exists.

Reproducibility

Raw JSON, generated tables and plots, environment audit, and runners are under `docs/papers/shared/results/paper6_3_openvino` and `experiments/paper6_3_openvino`. The version matrix is produced one process per cell by `run_cache_lifecycle_cell.py`; `summarize_cache_lifecycle_versions.py` parses the JSON and teardown logs into Table 5.

References

- [1] Intel. *OpenVINO GenAI LLMPipeline and SchedulerConfig API Reference*. https://docs.openvino.ai/2026/api/genai_api/
- [2] OpenVINO Toolkit. *OpenVINO GenAI*. <https://github.com/openvinotoolkit/openvino.genai>